



Figure 1: The basic steps to ensure accessibility in mobile apps

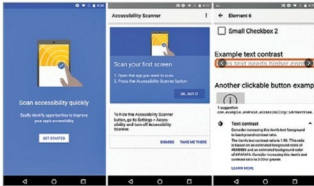


Figure 2: Accessibility scanner

Operability

The major factors related to operability are listed below.

Keyboard support: Though the mobile apps are primarily navigated using the touch screen inputs, maintaining compatibility with physical keyboards and equivalent accessibility devices is an important consideration. These keyboard-compatible interfaces are very useful for those with visual disabilities, and persons with dexterity related issues. The *focus order* of the controls needs to be maintained in a logical sequence, which makes the apps more accessible.

Touch target size and spacing: Maintaining operable sizes and adequate spacing between touch targets are two important considerations. The recommended minimum size for a touch target is 9mm x 9mm. In case of smaller-sized touch targets, the surrounding spaces need to be inactive in order to avoid touch overlap.

Gestures: In cases of complex gestures, design alternatives should be provided for screen readers, head pointer users, etc.

Understandable

The factors that impact 'understandability' are listed below.

- Instructions for gestures and other interaction methods need to be provided in an easily understandable manner.
- Maintaining standard shapes for controls, icons and links will improve the understandability of the interface.
- Placing critical user interface elements in positions that don't require scrolling is also an important factor.
- Avoiding language complexity is another factor in making the applications more understandable. It is always better to use simple terms and sentences.

Robustness

These factors need to be considered in order to make an app truly robust.

- Ensuring that the keyboards match the type of data to be entered is an important step. Even marginal customisation to the standard keyboard is confusing to screen-reader users.
- The characteristic properties of the platform need to be harnessed to make the applications robust.

Table 1: Factors impacting mobile accessibility

A11Y dimensions	Major factors
Perceivable	- Screen size considerations - Magnification/Zoom settings - Contrast levels
Operable	- Keyboard support - Touch target size - Gesture management
Understandable	- Understandable language - Instructions for gestures - Placing critical UI controls before scroll - Maintaining standard shapes for controls, icons, etc
Robust	- Keyboard standardisations - Utilising platform characteristics

Basic steps to make an Android app accessible

The following are the basic steps that make an Android app accessible.

Step 1: The user interface controls in the application need to be provided with a descriptive text. The descriptive text should be added with the `android:contentDescription` attribute. The text added through `contentDescription` is not to be displayed on the screen. However, this enables the audio prompt of the text when the application is accessed with accessibility tools.

For example, the following code snippet adds the content description 'Add Comment' to an `ImageButton` button:

```
<ImageButton
    android:id="@+id/add_comment"
    android:src="@drawable/add_comment"
    android:contentDescription="@string/add_comment"/>
```

In the case of text boxes, the attribute `android:hint` should be added. This attribute is used to provide a *hint* to the user regarding the content to be entered in the text box. After the content is entered, the hint is discarded and the text entered in the text box is read out by screen readers.

Step 2: The user interface controls need to be made navigable through physical keyboards and devices such as directional pads, eyes-free keyboards, etc. This 'focus navigation' plays an important role in making the application accessible. Focus management for user interface controls should be carried out by setting the `android:focusable` attribute to `True`. The methods `setFocusable()`, `isFocusable()` and `requestFocus()` will be used to set focus and to test focus.

The order of focus must be managed with the attributes listed